

 Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Electrical Engineering In-Semester Examination- II Academic Year: 2024-25		
Class: Final year (B-Tech)	Subject: Power Quality and Harmonics	Subject Code – PEC-EE414
Date: 10/10/2025	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- i. All Questions are Compulsory
- ii. Figures to the right indicate full marks.

Q.N.	Questions	Marks	CO	BT
Q.1	Explain Signal Reference Grid (SRG)? and its role in controlling noise and improving grounding performance.	5	CO3	BT2
Q.2	Compare single-point grounding and multi-point grounding systems used in PQ measurement setups.	5	CO3	BT2
Q.3	Explain Total Harmonic Distortion (THD) and Distortion Factor with suitable formulas.	5	CO4	BT2
Q.4	How do harmonics affect the accuracy of power and energy measurements in electrical systems?	5	CO4	BT1

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO 1	Understand definitions of power quality, power quality standards
CO 2	Distinguish between voltage sag and swell.
CO 3	Identify power quality disturbances & classify power quality problems.
CO 4	Understand the methods to mitigate harmonics in system.
CO 5	Design Active and Passive filters.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Electrical Engineering
In-Semester Examination II
Academic Year: 2025-26 (semester I)

Class: B.Tech.

Course: High Voltage Engineering

Course Code: PEC-EE405

Date: 09.10.2025

Time: 14:30 PM to 15:15 PM

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to right indicate full marks.
- The symbols carry usual meaning.

Q.N.	ATTEPT ANY FOUR	Marks	CO	BT
Q.1	Explain the phenomenon of breakdown due to internal discharge in solid dielectrics with neat diagram and equations.	5	CO3	BT2
Q.2	Explain the effect of moisture content on breakdown strength of liquid dielectric and stressed oil volume theory.	5	CO3	BT2
Q.3	Explain the phenomenon of Thermal Breakdown in dielectric materials. Derive the condition for thermal instability.	5	CO3	BT2
Q.4	With neat diagram, explain the construction and working of a multistage impulse generator (Marx circuit).	5	CO3	BT2
Q.5	Explain with neat circuit diagram the generation of impulse current waveform using capacitor bank and series RLC circuit.	5	CO3	BT2
Q.6	Explain with neat diagram the construction and working of a Tesla Coil used for generation of high frequency AC high voltage.	5	CO4	BT2

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Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Observe the breakdown mechanism in gaseous, liquid and solid insulation.
CO 2	Illustrate the method of high voltage generation, impulse voltage and current.
CO 3	Describe the methods of Measurement High voltage, Impulse voltage and current.
CO 4	Design various circuits for the measurement of high frequency voltages and currents.
CO 5	Apply the different tests done on insulators, circuit breakers, cables, transformers LA etc.



**Shri Guru Gobind Singhji Institute of Engineering and Technology,
Nanded
Department of Electrical Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)**

Class: Final Year	Subject: Switchgear and Protection	Subject Code: PCC EE 402
Date: 13.10.2025	Time: 14:30 to 15:45	Max. Marks: 20

Note:

- i. **All Questions are compulsory.**
- ii. *Mobile phones and programmable calculators are strictly prohibited.*
- iii. *Exchange/Sharing of stationery, calculator etc. not allowed.*
- iv. *Figures to right indicate full marks.*
- v. *Question paper contains only 01 pages.*

Q.N.	Questions	Marks	CO	BT
Q.1	Differentiate between impedance, reactance, and mho distance relays	6	CO 3	BT 2
Q.2	Describe the construction and working of a directional over-current relay	4	CO 3	BT 2
Q.3	What is pilot wire protection? Explain its working in transmission system protection	6	CO 4	BT 1
Q.4	State the philosophy of relay coordination and write steps to coordinate relays in interconnected systems .	4	CO 4	BT 1

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Knowledgeable in field of power system protection, circuit breakers, protective relaying and instrument transformer
CO 2	Comprehensive study of various relays used in power system protection
CO 3	Discuss types of circuit breakers with their applications
CO 4	Identify rotor ,stator faults, interterm faults and their protection
CO 5	Design relevant protection system for the power system